

RANKERS BATCH

NEET

Botany

Cell cycle and cell division

one shot

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Basics of cell cycle - Every organism Derive From Single cell, Cell Division & Growth is feature of Cell (Cell Growth → Divide into two Daughter)
 (From Zygote Division)
 Cell cycle → Coordinate Event → Cell Growth, DNA Double Divide & Distribute → intact genome two Daughter cell
 (Specific Phase)

Feature of cell cycle → Coordinated ✓
 → Genetic control ✓
 → One phase DNA Double, One phase Genome distribute into two cell

(Eukaryotic cell)

Cell cycle



(two phase) → Phase

①

M phase

- cell Division

- 5% time

- Karyokinesis - nucleus Divide
 Cytokinesis → Cytoplasm

② Interphase

→ B/w two Successive M-phase

- 95% time of cell cycle

- Resting phase → cell prepare For Division

→ $G_1 + S + G_2 = \text{Interphase}$

Gap-2 (Gap B/w S & M-phase)



Gap 1
 B/w M & S

Topics - ① Time

② Time

→ Yeast cell - 90 min Human cell - 24 hr

(Vary → two cell of same organism (culture)

- two cell of different organism

$G_1 > S > G_2 > M$
 $8-12 \text{ hr} > 6-8 \text{ hr} > 3-4 \text{ hr} > 1 \text{ hr}$ - Cell Divide 24 hr

③ Changes in phase

(transcription, translation)

G_1 → Post Mitotic, Gap-1 Growth in cytoplasm/cell occur

three option - ① Cell Become permanent ② Enter into S-phase

③ Exit G_1 → Enter G_0 (quiescent) - cell living, metabolically Active (not Proliferate)

S - Synthetic (DNA Double in Nucleus)
 (Histone synthesis)

Centriole Double in cytoplasm - Animal cell

G_2 → Gap-2 - cell prepare For Division (Pre mitotic)

Protein synthesis → tubulin → Give Microtubule

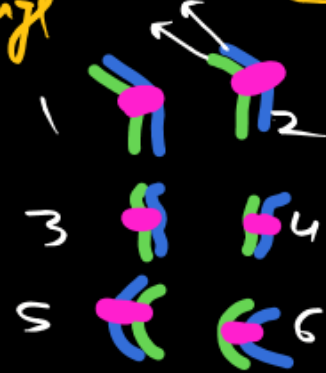
(Mitochondria Plastid Double in G_2)

III Amount change Ploidy change

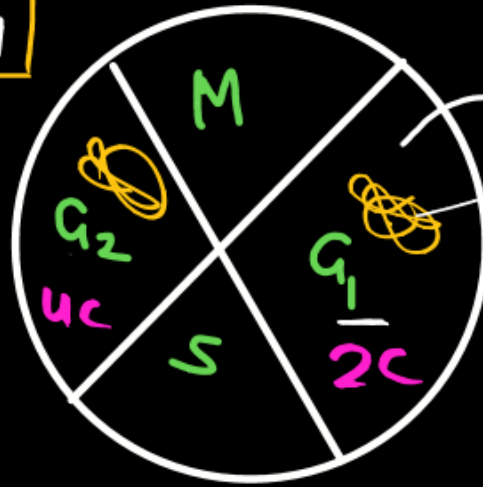
→ Cell cycle → number of chromosome - not change

Chromosome = 6
Chromatid = 12
Centomere = 6

1 chromosome
2 chromatid



Diploid cell

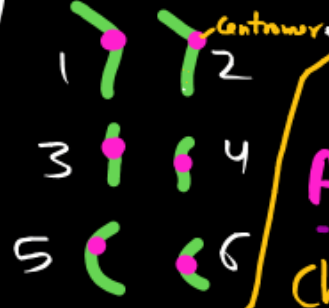


(Example)

total chromosome = 6

Chromatid → 6

Amount → 2c (Amount of DNA of chromatin)



	G ₁	S	G ₂	M
DNA Amount	2c	2c → 4c	4c	4c → 2c
Chromosome	6	6	6	6
Chromatid	6	Change	12	6

Each Daughter cell

(Amount of DNA change)

→ Amount change in phase

→ Number of chromatid change

→ Number of chromosome change

S phase
(2c → 4c)

M-phase
(4c → 2c)

G₁ phase - 6 chromatid / chromosome

G₂ phase - 12 chromatid / 6 chromosome

→ NO

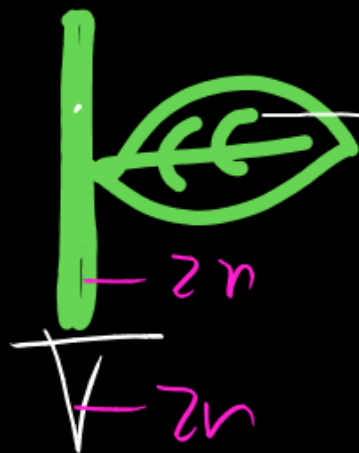
Diploid + G₁ - 2c ✓

G₂ = 4c ✓

haploid → c (Amount of DNA)

if Haploid Divide

G₁ - c G₂ = 2c



(Vegetative cell / Somatic cell)

Leaf cell → 24 chromosome

(Diploid cell)

Chromosome in Root, Stem = 24 chromosome

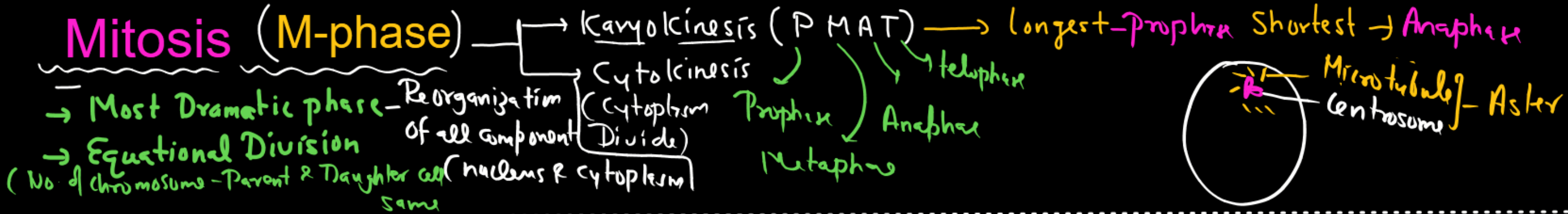
Chromosome of in G₁ → 24 Any cell

Chromatid in G₁ → 24
G₂ → 48

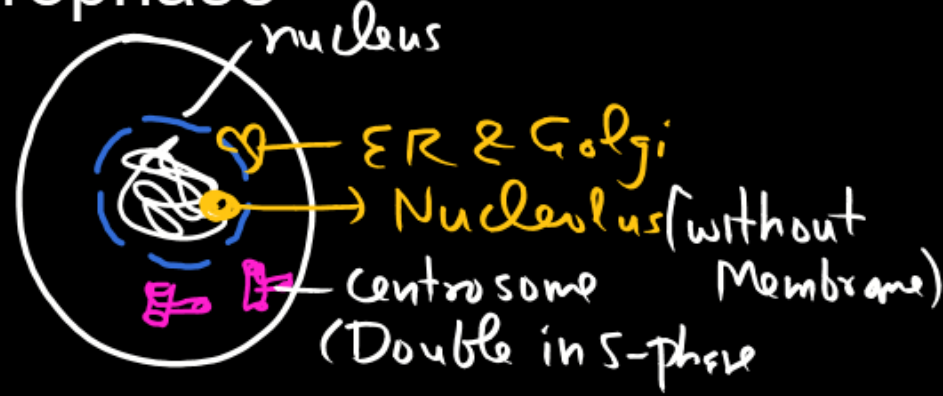
Chromosome in Pollen mother cell (2n) → 24

Amount of DNA → (2n) Pollen → n → 12
PMC → G₁ & G₂ → 4c
Pollen → c → 2c

Mitosis (M-phase)



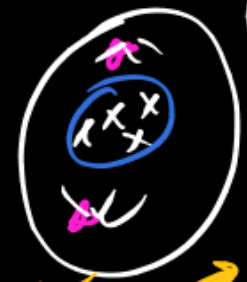
Prophase



Chromatin → Still → Untangling of Double (S-phase) intertwined Chromatin (3rd & 4th)

Chromatin Condense → Chromosome (not Fully) Visible

Early Prophase



Centrosome move opposite pole
Spindle Form

Centrosome + Microtubule = Aster
two Aster + spindle = spindle Apparatus

ER, Nucleolus, Golgi } Disappear
Nuclear Membrane start to Disintegrate

Mid Prophase Late Prophase

Metaphase →



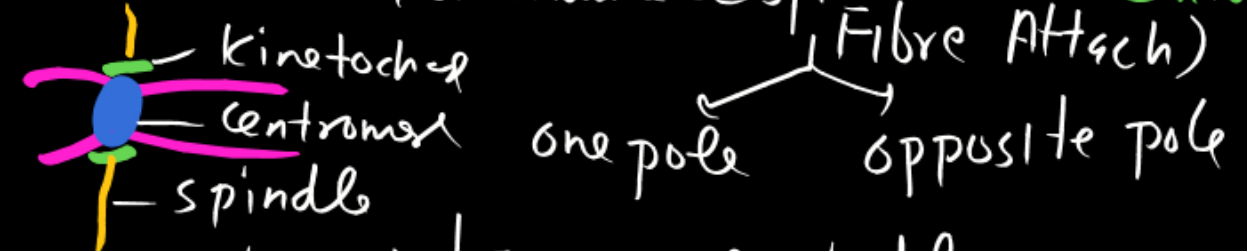
First Event - nuclear Membrane Break (completely)

Chromosome come in cytoplasm
Chromosome Fully condensed - Study of Morphology possible



Spindle Fibre attach on Kinetochore

In Such a manner → 1 chromosome = 2 spindle



Chromosome + come in centre / Equator of spindle
Plane of Alignment of chromosome at Equator = Metaphase plate

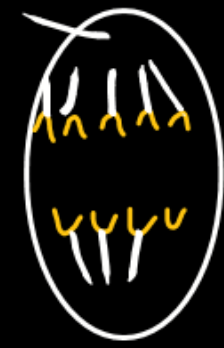


Anaphase

→ All chromosome - split simultaneously → Sister chromatid separate



Metaphase



Chromosome
Sister Chromatid

Chromosome
Future Daughter cell

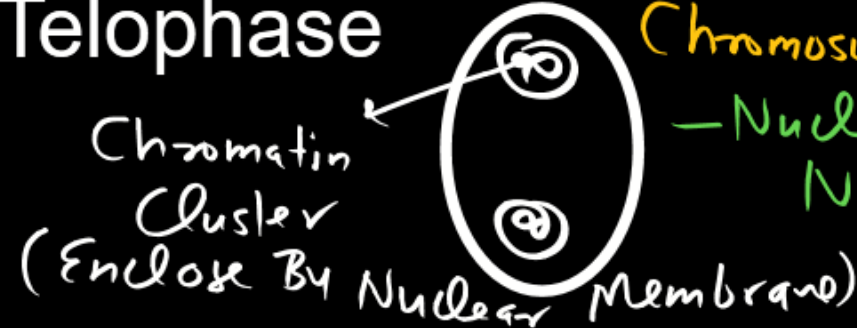
→ Sister chromatid is now Daughter Chromosome, moving opposite pole

→ Centromere Directed towards pole & Arms trailing Behind

Significance (Mitosis)

- (a) Genetically identical cells
- ✓ (b) nucleus/cytoplasm Ratio Restore (Growth - n/c dec) Cell division - n/c inc
- ✓ (c) Growth, Repair
- ✓ (d) occur in Diploid cells in Animals (Exception - Social insect - drone Honey bee) Mitosis in haploid
- (e) occur in plants → Diploid & Haploid (Location → Gut, Epidermis, Blood cells (lower plants) Meristems)

Telophase



Chromatin Cluster (Enclosed By Nuclear Membrane)
Chromosome - decondense (no longer individual chromosome)
Nuclear Membrane, ER, Golgi
Nucleolus → Reappear
Identity lost

Cytokinesis - Cytoplasm & Organelles - Distribute in Daughter cells

(Animal cell) - No cell wall
(a) Cleavage Method → Furrow → deepens → Meet in centre

(Plants → Rigid cell wall)
(b) Cell Plate method → Plants → Middle Lamella deposit (inner to outer)
Calcium pectate (cell plate)

APC - Anaphase promoting complex

Sister Chromatid separate
Digest By APC → Cohesin

→ APC Functional in
If APC not Functional → Sister Chromatid Remain Attached (Polyteny)

Colchicine

→ (Chemical)
→ Spindle Formation stop
→ Sister chromatid separate → But Remain in same cell



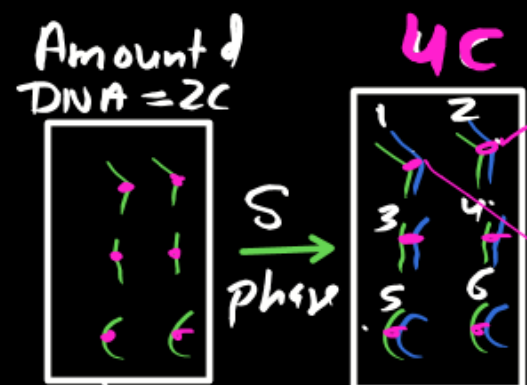
Chromosome Double (Polyploidy)

Meiosis

→ occur in Diploid cell
→ Why meiosis → gt produce haploid stage

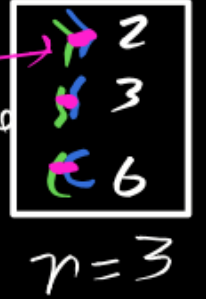
S-phase - one time?
Cell Divide → two time
Stages Meiosis I

Prophase - I
Metaphase - I
Anaphase - I
Telophase - I



S phase
Meiosis I

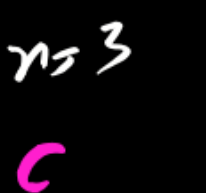
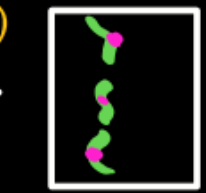
Reduction
(Chromosome No. half)



Meiosis II
P-II
M-II
A-II
T-II

Sister chromatid
Separate
(Like Mitosis)

Meiosis II



Feature

(Blw chromosome)

- gt give variation - Exchange of Genetic material
- Pairing of chromosome occur
- 4 haploid Daughter cell Form
- One cycle of DNA Replication, two cell Division
- Mitosis → Equational Division
- Meiosis → Reductional Division (Chromosome half)

Meiosis (Reductional)

In Meiosis I - Changes → Chromosome number half (Chromosome pair separate)
(Reduction)

In Meiosis II - Changes → Chromosome number constant, Amount half
(Equational)

Meiosis I

Prophase I - More complex, longer - Divided into phases on the basis of → Behaviour
(Compare Prophase - Mitosis)

L Z P D D

→ of chromosome

(A) **leptotene** → Chromatin condense
→ Condensation occur throughout



Chromosome visible (leptotene)

Zygotene → Pairing of chromosome occur → Process of pairing → **Synapsis**



Paired chromosome is called - **Bivalent (Pair) / tetrad**

Synaptonemal complex → help in pairing (visible in EM)

Human cell = 23 Bivalent or 23 tetrad

Pachytene - Bivalent More clearly visible - Pachytene

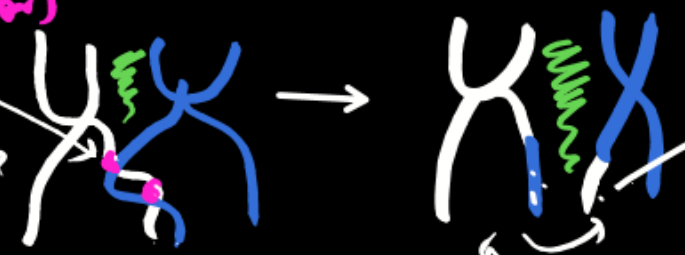
- Exchange of Genetic Material → B/w - **Non-sister chromatid**

- Recombination nodule (crossing over)
- Enzyme - Recombinase
- Result is Variation

(Exchange of Genetic Material)

Diplotene

→ **Synaptonemal complex** dissolution of
- Chromosome start to move away (some oocyte Arrest in this stage)
- Chiasmata visible



Exchange of Genetic material (Recombination)



cross like structure

Sister chromatids
non-sister chromatid

DIAKINESIS → terminalization of chiasmata

- Nuclear Membrane, ER, Golgi, nucleolus (Disappear)

- Chromosome fully condensed (This phase is transition to Metaphase I)



Metaphase I



Chromosome pair → Spindle Attach From opposite pole
1 pair = 2 spindle Fibre

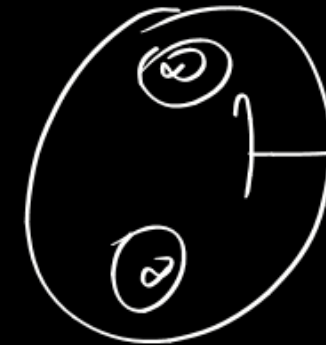
(Metaphase plate - Pair of chromosome)

Anaphase I - Pair separate



One chromosome From Each pair (move to opposite pole)

Telophase I → two pole + chromosome decondense → chromatin



two haploid nuclei form (ER, Golgi, nucleolus, nuclear membrane appear again)



Interphase

- Gap B/w two Mitosis
- DNA Replication occur in S-phase

Meiosis - II (like Mitosis)



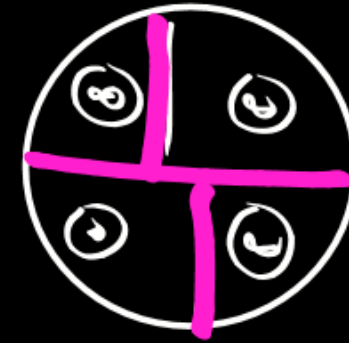
- Interkinesis → Gap B/w M-I & M-II
- No DNA Replication
- Centriole Double
- DNA not come in complete Dispersed state of Chromatin

Prophase II



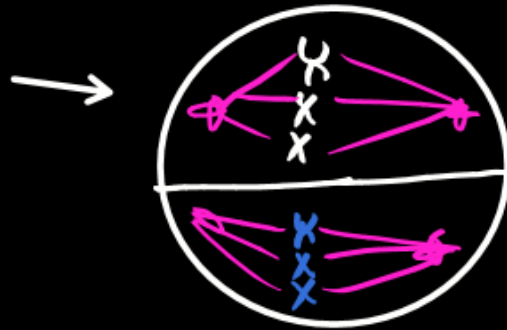
- Chromatin condense
- ER, Golgi, nucleus Disappear
- Nuclear Membrane Break

Telophase



- 4 haploid nuclei form
- Cytokinesis → 4 haploid cells (3)

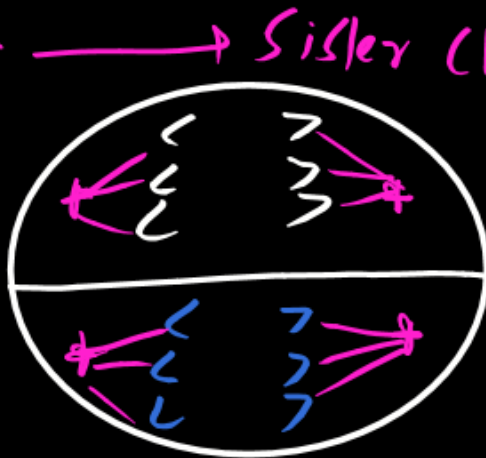
Metaphase II



- Each chromosome have two spindle Fibre - opposite pole

Anaphase II

(APC Functional)



- Sister Chromatid separate

Significance → Meiosis → Chromosome half (Haploid stage form) (But constant in species)
 → Variation By - Crossing over → Helpful in Evolution